

7. FCI Space Explosion and EWF's Scaling Advantage

FCI dimension $\binom{2n}{N_e}$ grows exponentially. EWF splits into F fragments, each with n_{frag} orbitals and N_{frag} electrons, solved independently.

Given: Test molecules (a)–(e). C_2H_4 's EWF max fragment: 20 qubits ($n_{\text{frag}} = 10$).

Solve:

- (a) Compute FCI dimensions for all 5 molecules, tabulate, observe trend.
- (b) Prove: fixed n_{frag} , $F = N/n_{\text{frag}} \rightarrow$ EWF cost $O(N)$.
- (c) Compute C_2H_4 max fragment FCI dimension, compare with full $\sim 3.1 \times 10^7$.
- (d) Derive crossover: direct FCI $O(\binom{2N}{N_e}^3)$ vs. EWF $F \times O(\binom{2n_{\text{frag}}}{N_{\text{frag}}}^3)$. At what n_{frag} are they equal?

Advanced Challenge: Under 20-qubit NISQ constraint ($n_{\text{frag}} \leq 10$), optimal n_{frag} for N_2 and C_2H_4 ?

Solution:

- (a) 直接计算即可, H_2 是 $\binom{4}{2} = 6$, LiH 是 $\binom{8}{4} = 70$, H_2O 是 $\binom{14}{10} = 1001$, N_2 是 $\binom{20}{14} = 38760$, C_2H_4 是 $\binom{28}{16} = 30421755$
- (b) 当 n_{frag} 固定时, 每个片段的 FCI 维数 $\binom{2n_{\text{frag}}}{N_{\text{frag}}}$ 有上界, 其对角化代价为 $O(D_{\text{frag}}^3) = O(1)$, 片段总数为 $F = N/n_{\text{frag}}$, 所以 EWF 总代价为 $O(N)$
- (c) C_2H_4 最大碎片为 20 量子比特, 该碎片的 FCI 的维数最大是 $\binom{20}{10} = 184756$, 全分子的 FCI 的维数最大是 30421755, 二者之比

$$\frac{184756}{30421755} \approx 0.6\%$$

- (d) 假设电子均匀地分布在每个碎片中, 得到方程

$$\binom{2n_{\text{frag}}}{N_e n_{\text{frag}}/N} = \binom{2N}{N_e} \left(\frac{n_{\text{frag}}}{N}\right)^{1/3}$$

我们来证明: 对于 $n_{\text{frag}} < N$, 左边总是比右边小很多倍, 进而 EWF 总是比直接的 FCI 需要更少的时间复杂度 (显然当 $n_{\text{frag}} = N$ 时是相等的)

设 $\frac{N}{n_{\text{frag}}} = t$, 把左边的组合数记为 $\binom{a}{b}$, 有如下推导:

$$\binom{2N}{N_e} = \binom{ta}{tb} = \prod_{i=1}^{tb} \frac{ta - i + 1}{tb - i + 1} = \prod_{i=1}^n \frac{ta - i + 1}{tb - i + 1} \cdot \binom{a}{b} \geq (ta - tb + 1) \binom{a}{b} > \sqrt[3]{t} \binom{a}{b}$$

证明完毕

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